

4(a)	1 covalent / peptide ; 2 hydrogen ; 3 ionic / electrostatic ; 4 disulfide ; 5 hydrophobic interactions ; 6 Van der Waals ; <i>4 correct = 2 marks</i> <i>2 / 3 correct = 1 mark</i>	2
4(b)	remove sugar phosphates from, active site / rubisco / enzyme ; A breaks down sugar phosphate	1
4(c)(i)	<i>any three from:</i> 1 RuBP and rubisco in bundle sheath cells ; 2 mesophyll cells surround the bundle sheath cells ; 3 stops, air / oxygen, getting to bundle sheath cells ; 4 (so) oxygen does not, combine with RuBP / react with rubisco or (so) carbon dioxide, combines with RuBP / reacts with rubisco ;	3

Question	Answer	Marks
4(c)(ii)	<i>any two from:</i> 1 increase in (rate of), light dependent stage / photophosphorylation / photolysis ; 2 (so) increase in oxygen produced ; 3 leads to an increase in oxygen to carbon dioxide ratio ; 4 favours reaction with oxygen (ref. to rubisco) ; 5 more stomata open ;	2
4(c)(iii)	<i>any two from:</i> 1 less RuBP to combine with carbon dioxide / less carbon fixation ; 2 less, GP / TP ; 3 ref. to reduction in (rate of), Calvin cycle / light independent stage ;	2